Introduction

This document provides a detailed explanation of the manual annotation protocol used for the TXL-PBC dataset. The TXL-PBC dataset is constructed by integrating four different publicly available blood cell datasets: BCCD, BCDD, PBC, and Raabin-WBC. The detailed sources and references for these datasets can be found in our main article.
We manually selected 250 images from the four datasets for initial manual labeling: 100 images from the PBC dataset, 100 images from the Raabin-WBC dataset, 30 images from the BCCD dataset, and 20 images from the BCDD dataset.
This manual annotation protocol outlines the standards and guidelines we followed to ensure high-quality and consistent annotations throughout the process.

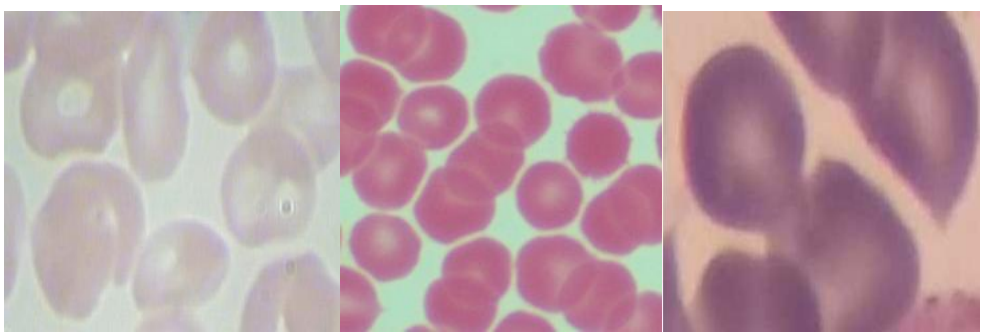
Annotation Tool Description

For manual annotation, we used the open-source tool X-AnyLabeling. X-AnyLabeling is a versatile graphical image annotation tool that supports multiple annotation formats, including the YOLO format used in our project. Using its intuitive interface, annotators can easily open image folders, select the desired annotation format, and draw bounding boxes around each cell of interest by clicking and dragging. Each bounding box can be assigned the correct label (e.g., RBC, WBC, Platelet), and the annotation files are saved in the YOLO format in the same directory as the images. X-AnyLabeling also provides convenient keyboard shortcuts to improve annotation efficiency.
For more details, refer to the official documentation at:
https://github.com/CVHub520/X-AnyLabeling.

Definition of Annotation Targets

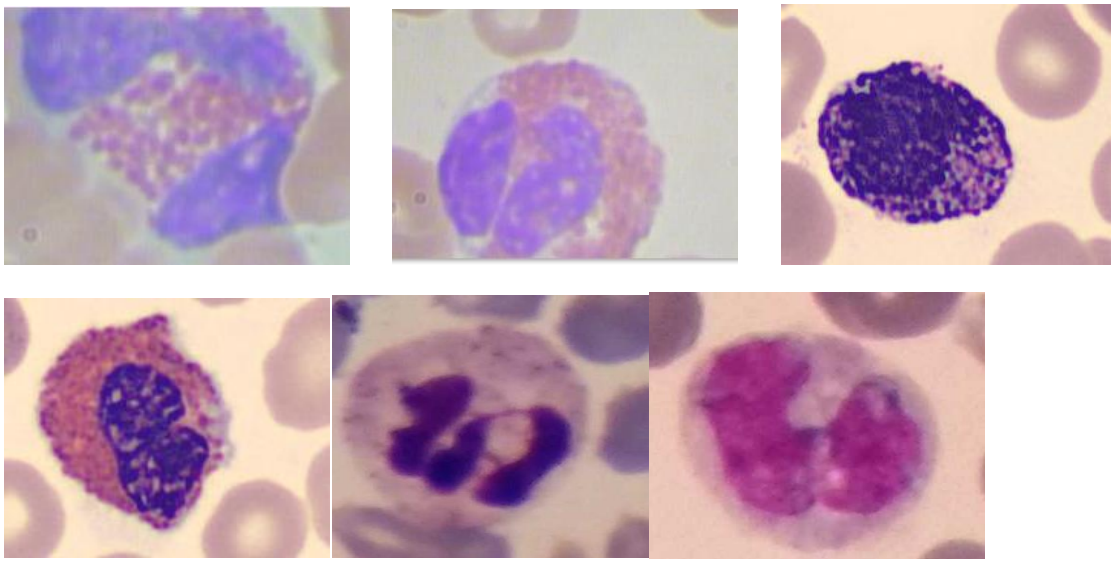
Red Blood Cell (RBC)

Definition: Red blood cells are the most abundant cells in peripheral blood. They are typically round, biconcave, and lack a nucleus.
Identification Criteria:
Uniform, circular or oval shape
Central pale area (biconcave appearance)
No visible nucleus
Stains pink or light red in standard blood smears



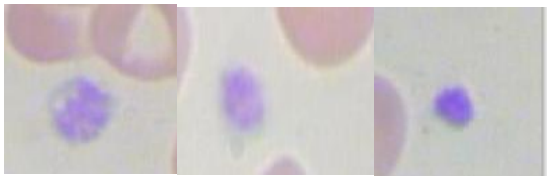
White Blood Cell (WBC)

Definition: White blood cells are immune cells present in blood, larger than RBCs, and contain a nucleus. There are several subtypes, but for annotation, all are labeled as WBC.
Identification Criteria:
Larger than RBCs
Presence of a clearly visible nucleus (may be segmented or round)
Cytoplasm may be granular or clear
Stains purple or blue in standard blood smears



Platelet

Definition: Platelets are small, anucleate cell fragments involved in blood clotting.
Identification Criteria:
Much smaller than RBCs and WBCs
Irregular or round shape
No nucleus
Stains light purple or blue



Annotation Procedure

Open Images

Launch LabelImg and open the folder containing our images.

Select Format and Category

Choose the YOLO format. Select or enter the correct cell type label (RBC, WBC, Platelet).

Draw Bounding Boxes

Use the mouse to draw a tight bounding box around each cell. Make sure the box closely fits the cell boundary.

Save Annotations

After labeling all cells in the image, click “Save” to store the annotation file in YOLO format.

What to Annotate

Annotate all clearly visible RBCs, WBCs, and Platelets.

Skip cells that are too blurry or cannot be confidently identified.

For overlapping cells, draw separate boxes if boundaries are clear.

For edge cells, annotate only if more than half is visible.

Special Cases

Only annotate blurry or poorly defined cells if we are confident about their type.

Do not annotate if the cell cannot be reliably separated from others or the background.

Quality Check

Review all boxes before moving to the next image to ensure accuracy and consistency.

Quality Control

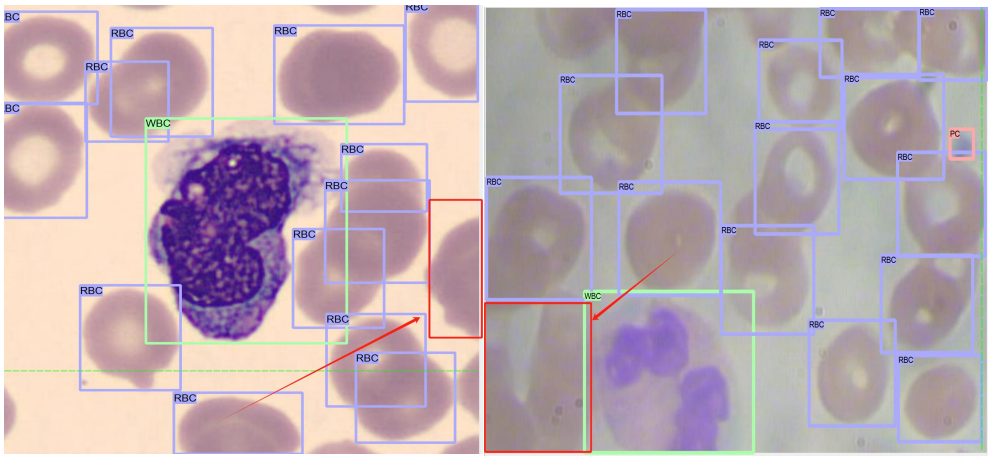
After completing the annotation for each image, the annotator performs a self-check by reviewing all bounding boxes to ensure that each cell is correctly labeled and that the bounding boxes are accurately placed. Any obvious mistakes or omissions are corrected immediately. Once the self-check is complete, a second annotator independently reviews the same set of images (cross-check). The second annotator checks for consistency in cell type labeling, the accuracy of bounding box placement, and whether any cells have been missed or incorrectly annotated.

For cells that are difficult to identify or have low confidence (e.g., blurry, overlapping, or partially visible cells), we only label the cell if we are confident about its type. In our annotation process, If the cell type cannot be determined with reasonable certainty, it is left unannotated. Only cells we were confident about were included in the final annotations. This approach helps to ensure the overall quality and reliability of the dataset.

Suggestions for Handling Common Problems

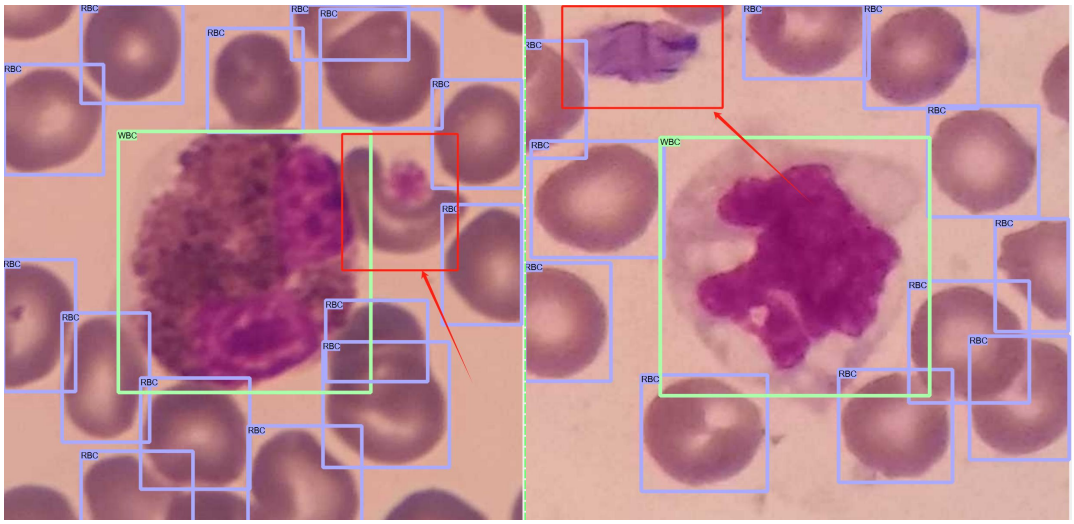
Overlapping Cells:

When two or more cells overlap, draw a separate bounding box for each cell as long as their boundaries can be reasonably distinguished. If the overlap is so severe that the individual cells cannot be separated, do not annotate that region.



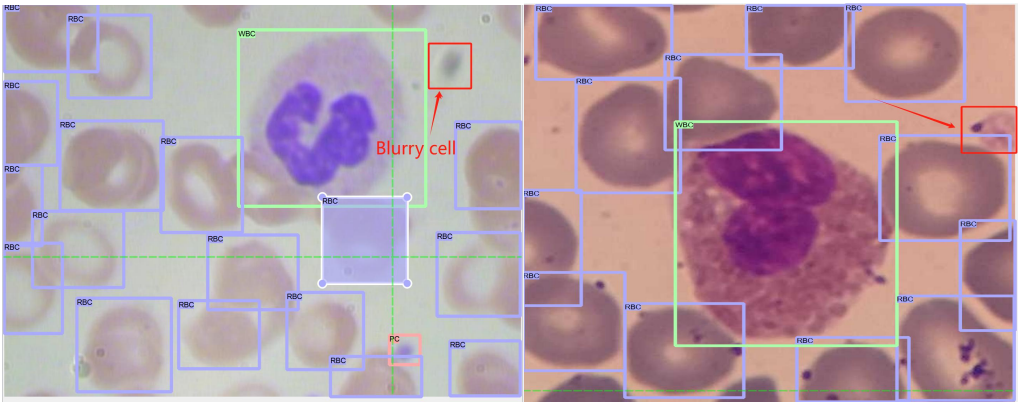
Damaged Cells:

If a cell is partially damaged but its type can still be confidently identified, annotate it as usual. If the damage is too severe to determine the cell type, skip annotation for that cell.



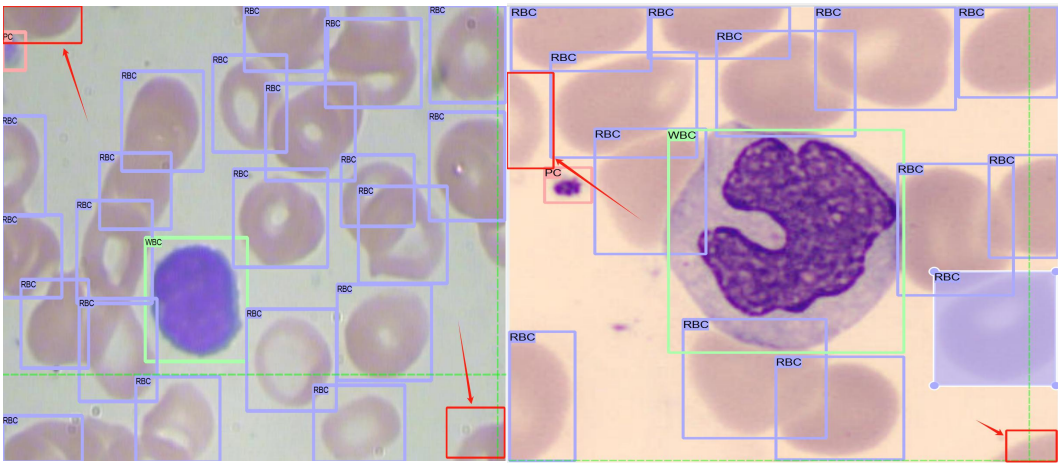
Blurry Images or Cells:

Only annotate cells that can be confidently identified, even if the image or cell is slightly blurry. If the cell is too blurry to determine its type, do not annotate it. For images where most cells are blurry and cannot be reliably identified, consider excluding the entire image from annotation.



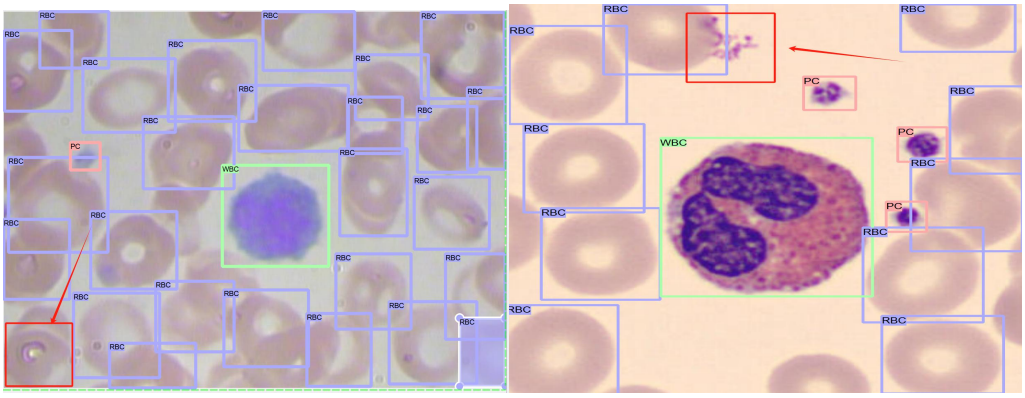
Cells at the Edge of the Image:ss

Annotate cells at the edge only if more than half of the cell is visible and its type can be confidently determined. Otherwise, skip annotation for these cells.



Unusual Cell Morphology:

If a cell appears abnormal but can still be classified as one of the target types, annotate it according to its most likely type. If the morphology is too ambiguous, do not annotate.



These guidelines help ensure that only high-confidence, clearly identifiable cells are included in the dataset, improving the overall quality and reliability of the annotations.